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Inventor(s)	Liu; Li-Shi

Temperature-controlled plasma generation system

Abstract

The present disclosure relates to a plasma generation system with a dielectric window, an inductive coil disposed on the dielectric window, a gas distribution element disposed on the dielectric window, and a gas conditioning system coupled to the gas distribution element. The gas distribution element is configured to discharge a thermally conditioned gas on the dielectric window and regulate a temperature across the dielectric window. The gas conditioning system is configured to supply the thermally conditioned gas to the gas distribution element.

Inventors:	Liu; Li-Shi (Taipei, TW)
Applicant:	Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)
Family ID:	1000008777882
Assignee:	Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 16/437,591, titled “Temperature-Controlled Plasma Generation System,” filed Jun. 11, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/691,916, titled “Dual Zone Heating System of Upper Electrode,” filed Jun. 29, 2018, each of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) With advances in semiconductor technology, there has been an increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices. Such scaling down has increased the complexity of semiconductor manufacturing processes and the demands for temperature regulation in semiconductor manufacturing systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional view of a semiconductor plasma processing system with a temperature-controlled plasma generation system, in accordance with some embodiments.

(3) FIG. 2 is a graph illustrating temperature uniformity of a dielectric window of a temperature-controlled plasma generation system, in accordance with some embodiments.

(4) FIGS. 3A-3B illustrate a top view and a cross-sectional view, respectively, of a temperature-controlled plasma generation system, in accordance with some embodiments.

(5) FIG. 4 illustrates an isometric view of a temperature-controlled plasma generation system, in accordance with some embodiments.

(6) FIGS. 5A-5B illustrate isometric views of gas conditioning systems of a temperature-controlled plasma generation system, in accordance with some embodiments.

(7) FIG. 6 illustrates an exploded isometric view of a gas distribution element of a temperature-controlled plasma generation system, in accordance with some embodiments.

(8) FIGS. 7A-7B illustrate a back side view and a cross-sectional view, respectively, of a cover plate of a gas distribution element of a temperature-controlled plasma generation system, in accordance with some embodiments.

(9) FIGS. 8-9 illustrate back side views of a base plate of a gas distribution element of a temperature-controlled plasma generation system, in accordance with some embodiments.

(10) FIG. 10 is a flow chart of a method for regulating and maintaining a temperature across a dielectric window of a temperature-controlled plasma generation system, in accordance with some embodiments.

(11) Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(15) It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

(16) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value).

(17) Semiconductor wafers are subjected to different plasma processes (e.g., plasma etching and/or plasma enhanced deposition) in different plasma processing systems during the fabrication of semiconductor devices. The plasma processing systems provide temperature-controlled plasma processing chambers to prevent temperature dependent process drifts during wafer plasma processing to achieve wafer-to-wafer reproducibility for low manufacturing costs and high production yield of semiconductor devices.

(18) In the plasma processing systems, the temperature dependent process drifts can occur due to temperature variations in plasma generation systems. As the plasma generation systems are positioned on the plasma processing chambers, temperature variations in the plasma generation systems can cause the properties of the plasma generated in the plasma processing chambers to vary, consequently causing variations in plasma process parameters (e.g., etch rate and/or deposition rate) associated with the processed wafers. For example, the temperature of a dielectric window of a plasma generation system can increase with increase in the number of plasma processed wafers. As the bottom surface of the dielectric window forms the top surface of the plasma processing chamber, the temperature variations in the dielectric window can induce variations in the chemical composition of the plasma generated in the plasma processing chamber. As a result, the plasma process parameters (e.g., etch rate and/or deposition rate) can vary between the wafers processed in the plasma processing chamber and poor wafer-to-wafer reproducibility.

(19) The present disclosure provides example temperature-controlled plasma generation systems for plasma processing systems to prevent and/or mitigate temperature dependent process drifts during wafer plasma processing to achieve wafer plasma process repeatability. In some embodiments, a temperature-controlled plasma generation system can include a dielectric window and one or more gas distribution elements positioned on the dielectric window. The one or more gas distribution elements can be configured to distribute thermally conditioned gas on the dielectric window and regulate the temperature across the dielectric window to be substantially uniform and constant during wafer plasma processing.

(20) In some embodiments, the one or more gas distribution elements can have ring-shaped tubular structures and a plurality of gas distribution holes configured to discharge thermally conditioned gas onto a top surface of the dielectric window. In some embodiments, different regions of the one or more gas distribution elements can have different sets of gas distribution holes. The dimension and density of each set of gas distribution holes can be different from each other. The sets of gas distribution holes can be designed and arranged to control the thermally conditioned gas amount discharged across the dielectric window to apply a substantially uniform amount of thermal energy across the dielectric window and regulate the dielectric window temperature to be substantially uniform during wafer plasma processing.

(21) In some embodiments, the dielectric window with the one or more gas distribution elements can have temperature variations less than about 6° C. (e.g., about 0.5° C., about 1° C., about 3° C., or about 5° C.) during wafer plasma processing. In some embodiments, the temperature variations in the dielectric window with the one or more gas distribution elements can be reduced by a range of about 80% to about 95% (e.g., about 80%, about 85%, about 90%, or about 95%) compared to the temperature variations in dielectric windows without the one or more gas distribution elements during wafer plasma processing. As a result, temperature dependent variations in plasma process parameters (e.g., etch rate and/or deposition rate) between plasma processed wafers can be reduced and plasma process repeatability and wafer-to-wafer reproducibility can be achieved.

(22) FIG. 1 illustrates a cross-sectional view of a semiconductor plasma processing system **100**, according to some embodiments. In some embodiments, semiconductor plasma processing system **100** can include a plasma processing chamber **102**, a temperature-controlled plasma generation system **104** (also referred to as “plasma generation system **104**) positioned on plasma processing chamber **102**, a wafer holder system **106** configured to hold a wafer **107** during plasma processing within plasma processing chamber **102**, and a control system **108** configured to control one or more operations of plasma generation system **104** and wafer holder system **106**. Plasma generation system **104** can be configured to generate plasma (not shown) within plasma processing chamber **102** for various plasma processes, such as etching or deposition.

(23) In some embodiments, plasma generation system **104** can include a dielectric window **110**, inductive coils **112A-112B** and gas distribution elements **114A-114B** on dielectric window **110**, gas conditioning systems **116A-116B** coupled to respective gas distribution elements **114A-114B**, sensors **118A-118B** on dielectric window **110** under respective gas distribution elements **114A-114B**, a gas injector **120** coupled to plasma processing chamber **102** through dielectric window **110**, and a radio frequency (RF) power supply **122** and a match network **124** coupled to inductive coils **112A-112B**.

(24) Plasma processing chamber **102** can be structurally defined by chamber sidewalls **102A**, base **102B**, and dielectric window **110**. Bottom surface **110b** of dielectric window **110** can form the top surface of plasma processing chamber **102**. Chamber sidewalls and base **102A-102B** can include stainless steel or aluminum and can be coated with yttrium oxide (Y.sub.2O.sub.3), yttrium fluoride (YF), cerium oxide (CeO.sub.2), and/or other suitable plasma resistant coatings. In some embodiments, plasma processing chamber **102** can include a gas inlet port **126** configured to receive one or more process gases (e.g., etchant gas or precursor deposition gas) from a process gas source (not shown) for the plasma processes. The one or more process gases and byproducts can be

removed from plasma processing chamber **102** through an outlet port (not shown) in base **102B** of plasma processing chamber **102** and a pump (not shown), which also can serve to maintain a particular pressure within plasma processing chamber **102**. In some embodiments, vacuum can be maintained in plasma processing chamber **102** during the plasma processing by a suitable vacuum pump apparatus (not shown) coupled to the outlet port.

(25) Dielectric window **110** can include a dielectric material, such as ceramic, quartz, aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), aluminum carbide (AlC), silicon (Si), silicon carbide (SiC), silicon nitride (SiN), or a combination thereof. In some embodiments, dielectric window **110** can have a vertical dimension **110t** (e.g., thickness) along a Z-axis ranging from about 2 cm to about 6 cm (e.g., about 2 cm, about 3 cm, about 4 cm, about 5 cm, or about 6 cm).

(26) Inductive coils **112A-112B** can each be positioned spirally around gas injector **120** on dielectric window **110**. Even though two inductive coils are shown in FIG. 1, plasma generation system **104** can have any number of inductive coils. Inductive coils **112A-112B** can include electrically conductive materials similar to or different from each other. Even though FIG. 1 shows each of inductive coils **112A-112B** to have three turns, each inductive coil can have any number of turns. Each of inductive coils **112A-112B** can be coupled to RF power supply **122** via match network **124**. RF power supply **122** can be configured to supply RF current to inductive coils **112A-112B** to create inductively coupled RF plasma in plasma processing chamber **102**. The RF current flowing through inductive coils **112A-112B** can generate an electromagnetic field about inductive coils **112A-112B**. The electromagnetic field can generate an inductive current within plasma processing chamber **102**. The inductive current can act on a plasma generating gas introduced into plasma processing chamber **102** through gas injector **120** to generate the inductively coupled RF plasma. RF power supply **124** can be configured to supply the RF power in a range of about 50 W to about 5000 W. In some embodiments, control system **108** can be configured to detect the mode of plasma generation based on the operation of the gas supply system (not shown) providing plasma generating gas through gas injector **120**. Plasma generation can be considered to be active when the gas supply system is in operation and can be considered to be terminated when the gas supply is off. In some embodiment, wafer **107** can be transferred out of plasma processing chamber **102** in response to control system **108** indicating the termination of the plasma generation.

(27) In some embodiments, match network **124** can be configured to tune the RF power supplied to inductive coils **112A-112B**. The RF power supplied to inner inductive coil **112A** can be equal to or different from the RF power supplied to outer inductive coil **112B**. The RF power supplied to inductive coils **112A-112B** can be tuned to control the plasma density in a radial distribution over the wafers processed in plasma processing chamber **102**. In some embodiments, the RF power supplied to inductive coils **112A-112B** can be tuned based on the processing parameters (e.g., etch rate or deposition rate) defined during the plasma processes. In some embodiments, match network **124** can be coupled to control system **108** and configured to operate based on control signals **108** received from control system **108**.

(28) Gas distribution elements **114A-114B** can be configured to receive thermally conditioned gas (e.g., air, an inert gas, or a combination thereof) from gas conditioning systems **116A-116B**, respectively and to discharge the thermally conditioned gas on dielectric window **110**. The thermally conditioned gas can help regulate the temperature across one or more surfaces (e.g., top surface **110a** and/or bottom surface **110b**) of dielectric window **110** to be substantially uniform and maintain the temperature at a value in the range of about 80° C. to about 110° C. (e.g., about 80° C., about 90° C., or about 100° C.) during wafer plasma processing in plasma processing chamber **102**. In some embodiments, dielectric window **110** can have temperature variations less than about 6° C. (e.g., about 0.5° C., about 1° C., about 3° C., or about 5° C.) during wafer plasma processing. As a result, variations induced in the chemical composition of the generated plasma in plasma processing chamber **102** due to dielectric window temperature variations can be substantially

mitigated compared to plasma generation systems without gas distribution elements and achieve substantially constant plasma process parameters (e.g., etch rate and/or deposition rate) during wafer plasma processing.

(29) Compared to plasma generation systems without gas distribution elements **114A-114B** on dielectric windows, temperature variations in dielectric window **110** can be reduced by a range of about 80% to about 95% (e.g., about 80%, about 85%, about 90%, or about 95%) during wafer plasma processing. FIG. 2 illustrates an example of such comparison between plasma processing systems with and without gas distribution elements similar to gas distribution elements **114A-114B** on dielectric windows. FIG. 2 shows plots **236** and **238** of dielectric window temperatures during sequential plasma processing of a series of wafers with respect to the wafer number in the series of wafers. Plots **236** and **238** represent dielectric windows of plasma processing systems with and without gas distribution elements, respectively. Comparison of plots **236** and **238** shows a substantially constant dielectric window temperature for the system with gas distribution elements, whereas the dielectric window temperature increases with increase in the number of wafers for the system without gas distribution elements.

(30) Referring back to FIG. 1, in some embodiments, gas distribution elements **114A-114B** can be ring-shaped tubes and can be positioned on dielectric window **110** in an alternating configuration with inductive coils **112A-112B** around gas injector **120** as shown in FIG. 1. Gas distribution elements **114A-114B** can have cross-sections of any geometric shape (e.g., rectangular, circular, elliptical, or polygonal) along their diameters.

(31) Gas distribution elements **114A-114B** can be laterally spaced apart from each other, from inductive coils **112A-112B**, and from gas injector **120**. In some embodiments, bottom surfaces **114Ab-114Bb** of gas distribution elements **114A-114B** can be vertically spaced apart from dielectric window **110** to create gaps **117** and allow thermally conditioned gas to be discharged from gas distribution holes (not shown in FIG. 1; shown in FIGS. 4, 6, and 7A-9A) in bottom surfaces **114Ab-114Bb**. Gaps **117** can each have a vertical dimension along a Z-axis ranging from about 1 mm to about 100 mm (e.g., about 1.5 mm, about 5 mm, about 10 mm, about 20 mm, or about 50 mm).

(32) In some embodiments, instead of the alternating configuration of FIG. 1, both gas distribution elements **114A-114B** can be positioned between inductive coils **112A-112B**, between gas injection **120** and inner inductive coil **112A**, or around outer inductive coil **112B**. In some embodiments, horizontal dimensions **114A.sub.w-114B.sub.w** of gas distribution elements **114A-114B**, respectively, along an X-axis can be equal to or different from each other. In some embodiments, horizontal dimensions **114A.sub.w-114B.sub.w** can range from about 10 mm to about 40 mm (e.g., about 10 mm, about 20 mm, about 30 mm, or about 40 mm).

(33) Gas conditioning systems **116A-116B** can be coupled to gas distribution elements **114A-114B** through gas inlet pipes **115A-115B**, respectively. Gas inlet pipes **115A-115B** can be coupled to top surfaces **114At-114Bt** of gas distribution elements **114A-114B**, respectively, through openings on top surfaces **114At-114Bt**. Gas conditioning systems **116A-116B** can be configured to receive a gas (e.g., air or inert gas), thermally condition (e.g., heat or cool) the gas, and supply the thermally conditioned gas to gas distribution elements **114A-114B** through gas inlet pipes **115A-115B**, respectively. In some embodiments, gas conditioning systems **116A-116B** can be configured to control the flow rate of the thermally conditioned gas supplied to gas distribution elements **114A-114B**.

(34) In some embodiments, gas conditioning systems **116A-116B** can be configured to supply the thermally conditioned gas at a substantially constant temperature ranging from about 70° C. to about 120° C. (e.g., about 70° C., about 80° C., about 90° C., about 100° C., about 110° C., or about 120° C.) during wafer plasma processing. In some embodiments, gas conditioning systems **116A-116B** can be configured to supply the thermally conditioned gas at different temperatures ranging from about 70° C. to about 120° C. (e.g., about 70° C., about 80° C., about 90° C., about

100° C., about 110° C., or about 120° C.) during wafer plasma processing. Gas conditioning systems **116A-116B** can be configured to dynamically adjust the temperature of the thermally conditioned gas supplied to gas distribution elements **114A-114B** during the plasma processing. The dynamic adjustment can be performed based on control signals (not shown) received by gas conditioning systems **116A-116B** from control system **108**. These control signals can be generated by control system **108** based on the temperatures of dielectric window **110** and/or discharged gas from gas distribution elements **114A-114B**. The temperature of the discharged gas from gas distribution elements **114A-114B** can be measured using sensors **118A-118B**, respectively. Sensors **118A-118B** can be placed on dielectric window **110** within gaps **117**. Even though two sensors shown in FIG. **1**, any number of sensors similar to sensors **118A-118B** can be placed on dielectric window **110** within gaps **117**. The temperature of dielectric window **110** can be measured using sensors **118A-118B** and/or **119**. Any number of sensors similar to sensors **119** can be placed on dielectric window **110**.

(35) Wafer holder system **106** can include a chuck **128** positioned within plasma processing chamber **102** and configured to hold wafer **107** during the plasma processing. Chuck **128** can include an electrostatic chuck. In some embodiments, chuck **128** can include electrostatic electrodes (not shown) configured to apply clamping voltage to chuck **128** for electrostatically clamping wafer **107** to chuck **107**. Chuck **128** can be electrically charged using an RF power supply **132** coupled to chuck **128** through match network **134**. RF power supply **132** can be tuned by match network **134** to supply power to the electrostatic electrodes. In some embodiments, the operations of RF power supply **132** can be controlled by control system **108**.

(36) In addition, semiconductor plasma processing system **100** can include other structural and functional components, such as RF generators, matching circuits, chamber liners, control circuits, actuators, power supplies, exhaust systems, etc. which are not shown for simplicity.

(37) FIG. **3A** shows a top view of a temperature-controlled plasma generation system **304** (also referred to as “plasma generation system **304**”) and FIG. **3B** shows a cross-sectional view of plasma generation system **304** along line A-A of FIG. **3A**, according to some embodiments. Elements in FIGS. **3A-3B** with the same annotations as elements in FIG. **1** are described above. The above discussion of plasma generation system **104** applies to plasma generation system **304** unless mentioned otherwise.

(38) Plasma generation system **304** can be positioned on plasma processing chamber **102** (partial structure shown in FIG. **3B**). Plasma generation system **304** can include dielectric window **110**, inductive coils **112A-112B** and gas distribution elements **314A-314B** on dielectric window **110**, gas conditioning systems **316A-316B** each coupled to gas distribution elements **314A-314B**, gas inlet pipes **315A-315** coupled to gas distribution elements **314A-314B** and gas conditioning systems **316A-316B**, and gas injector **120** coupled to plasma processing chamber **102** through dielectric window **110**.

(39) The above discussion of gas distribution elements **114A-114B**, gas conditioning systems **116A-116B**, and gas inlet pipes **115A-115B** applies to gas distribution elements **314A-314B**, gas conditioning systems **316A-316B**, and gas inlet pipes **315A-115B**, respectively, unless mentioned otherwise. Plasma generation system **304** can further include other structural and functional components similar to plasma generation system **104**, such as sensors **118A-118B**, RF power supply **122**, and match network **124**, which are not shown in FIGS. **3A-3B** for simplicity.

(40) In some embodiments, dielectric window **110** can be circular in shape and have a diameter ranging from about 45 cm to about 60 cm (e.g., about 45 cm, about 50 cm, or about 55 cm). Gas distribution elements **314A-314B** can be ring-shaped tubes placed around gas injector **120** in an alternating configuration with inductive coils **112A-112B** on dielectric window. In some embodiments, inner gas distribution element **314A** can have an outer diameter ranging from about 240 mm to about 270 mm (e.g., about 250 mm, about 256 mm, about 265 mm, or about 270 mm) and outer gas distribution element **314B** can have an outer diameter ranging from about 440 mm to

about 470 mm (e.g., about 450 mm, about 456 mm, about 465 mm, or about 470 mm). Diameter of outer gas distribution element **314B** can be smaller than the diameter of dielectric window **110**, in some embodiments.

(41) In some embodiments, horizontal dimensions **314A.sub.w-314B.sub.w** of gas distribution elements **314A-314B**, respectively, along an X-axis or a Y-axis can be equal to or different from each other. In some embodiments, horizontal dimensions **314A.sub.w-314B.sub.w** can range from about 10 mm to about 40 mm (e.g., about 10 mm, about 20 mm, about 30 mm, or about 40 mm). In some embodiments, gas distribution elements **314A-314B** can be spaced apart from each other along an X- or a Y-axis by a lateral distance **340** ranging from about 60 mm to about 90 mm (e.g., about 60 mm, about 70 mm, or about 90 mm) to prevent contact with each other and/or with outer inductive coil **114B**. Lateral distance **340** can depend on horizontal dimensions **314A.sub.w-314B.sub.w** and the size of outer inductive coil **114B**.

(42) Each of gas distribution elements **314A-314B** can be configured to receive thermally conditioned gas from gas conditioning systems **316A-316B** through gas inlet pipes **315A-315B**, respectively. Gas conditioning system **316A** can be configured to supply thermally conditioned gas to gas distribution elements **314A-314B** through gas inlet pipe **315A**, and gas conditioning system **316B** can be configured to supply thermally conditioned gas to gas distribution elements **314A-314B** through gas inlet pipe **315A**. Thus, each of gas distribution elements **314A-314B** can receive thermally conditioned gas at two different inlet ports (not shown), which are the inlet ports to which gas inlet pipes **315A-315B** are coupled to each of gas distribution elements **314A-314B**. The two inlet ports of each gas distribution elements **314A-314B** can be about 180 degrees radially apart from each other and help to provide a substantially uniform distribution of thermally conditioned gas with gas distribution elements **314A-314B**.

(43) The arrows on gas distribution elements **314A-314B** moving radially around a Z-axis indicate the flow of thermally conditioned gas through gas distribution elements **314A-314B** after being supplied through gas inlet pipes **315A-315B**. The arrows extending away from gas distribution elements **314A-314B** indicate the flow of thermally conditioned gas after being discharged from gas distribution elements **314A-314B** through their gas distribution holes on their base plates (not shown in FIG. 1; shown in FIGS. 6 and 8-9).

(44) FIG. 4 shows an isometric view of a temperature-controlled plasma generation system **404** (also referred to as “plasma generation system **404**”), according to some embodiments. Elements in FIG. 4 with the same annotations as elements in FIG. 1 are described above. The above discussion of plasma generation system **104** applies to plasma generation system **404** unless mentioned otherwise. Plasma generation system **404** can include a chamber **442**, dielectric window **110** and gas distribution elements **114A-114B** within chamber **442**, gas conditioning systems **116A-116B** coupled to respective gas distribution elements **114A-114B**, and gas inlet pipes **115A-115B** coupled to gas distribution elements **114A-114B** and gas conditioning systems **116A-116B**. Plasma generation system **404** can further include other structural and functional components similar to plasma generation system **104**, such as inductive coils **112A-112B**, gas injector **120**, sensors **118A-118B**, RF power supply **122**, and match network **124**, which are not shown in FIG. 4 for simplicity.

(45) Chamber **442** can include gas inlet and outlet ports **444A-444B** on its sidewalls **442A-442B** (ports **444A-444B** on sidewall **442A** not visible in FIG. 4). Gas inlet ports **444A** can be used to couple gas inlet pipes **115A-115B** to gas conditioning systems **116A-116B**, respectively. Gas outlet port **444B** can be used to extract all or portions of gas discharged by gas distribution elements **114A-114B** on dielectric window **110** within chamber **442**. Gas conditioning systems **116A-116B** can be configured to receive all or portions of the gas discharged within chamber **442** through gas outlet ports **444B**, thermally recondition the gas, and redistribute the gas on dielectric window **110** through gas distribution elements **114A-114B**.

(46) Gas conditioning system **116B** is further described with reference to FIGS. 5A-5B, which show isometric views of gas conditioning system **116B** with and without a housing **546**,

respectively, according to some embodiments. Gas conditioning system **116B** can include housing **546** with openings **548A-548B** and a gas inlet port **556**. Housing **546** can be coupled to sidewall **442B** of chamber **442** (FIG. 4) with openings **548A-548B** aligned with ports **444A-444B** (FIG. 4), respectively.

(47) Referring back to FIGS. 5A-5B, gas conditioning system **116B** can further include gas tubes **550A-550B**, a thermal conditioning apparatus **552**, a gas circulation system **554**, and a gas inlet port **558** within housing **546**. Gas tubes **550A-550B** can be coupled to housing **546** with openings **548A-548B** aligned with gas tube openings **560A-560B**, respectively, and gas inlet port **556** aligned with gas inlet port **558**. Gas conditioning system **116A** can also include elements similar to gas conditioning system **116B**.

(48) Gas to be thermally conditioned within gas conditioning system **116B** can be received by gas tube **550B** through gas inlet ports **556** and **558** and/or through gas tube opening **560B**. Gas received through gas tube opening **560B** can be the all or portions of gas extracted from chamber **442** through gas outlet port **444B**, which can be aligned to openings **548B** and **560B**. Gas circulation system **554** coupled to gas tubes **550A-550B** can be configured to draw the gas from gas tube **550B**, direct it through gas tube **550A** for thermal conditioning, and discharge the thermally conditioned gas through openings **560A** and **548A**. In some embodiments, gas circulation system **554** can include a suction fan configured to draw the gas and/or a blower for directing the gas. Thermal conditioning apparatus **552** disposed within gas tube **550A** can be configured to thermally condition (e.g., heat or cool) the gas received from gas tube **550B**. In some embodiments, thermal conditioning apparatus **552** can include a heating coil configured to heat the gas and/or a fan configured to cool the gas within gas tube **550A**. In some embodiments, gas conditioning systems **116A-116B** can represent gas conditioning systems **316A-316B**, respectively.

(49) FIG. 6 shows an exploded view of a gas distribution element **614**, according to some embodiments. Gas distribution element **614** can represent gas distribution elements **114A**, **114B**, **314A**, and/or **314B** discussed above with reference to FIGS. 1, 3A, and 4. In some embodiments, gas distribution element **614** can be ring-shaped and can include a cover plate **614A** with a gas inlet port **662** and a base plate **614B** with gas distribution holes **664**. Gas inlet port **662** can be coupled to a gas inlet pipe **115A**, **115B**, **315A**, and/or **315B** (not shown in FIG. 6; shown in FIGS. 1, 3A, and 4). Cover plate **614A** can have more than one gas inlet port. Gas distribution holes **664** can allow thermally conditioned gas within gas distribution element **614** to be discharged and distributed on dielectric window **110** (not shown in FIG. 6; shown in FIGS. 1, 3A, and 4). In some embodiments, gas distribution holes **664** can have dimensions similar to or different from each other. In some embodiments, gas distribution holes can have a circular shape as shown in FIG. 6 or can be any geometric shape (e.g., rectangular, elliptical, or polygonal).

(50) In an assembled form of gas distribution element **614**, base plate **614B** can be disposed on extended regions **668A-668B** along interior sidewalls **670A-670B** of cover plate **614A**. In some embodiments, in the assembled form, the top surfaces of recessed rim portions **672A-672B** of cover plate **614A** can be substantially coplanar with back surface **614B.sub.s** of base plate **614B** and the top surfaces of protruding rim portions **674A-674B** of cover plate **614A** can extend above back surface **614B.sub.s** along a negative Z-axis direction. Protruding rim portions **674** can be configured to support gas distribution element **614** on a dielectric window (e.g., dielectric window **110**) such that gaps (e.g., gaps **117**) between base plate **614B** and the dielectric window can be created to allow thermally conditioned gas to be discharged from gas distribution holes **664** as described with reference to FIG. 1.

(51) In some embodiments, exterior sidewalls **671A-671B** of cover plate **614A** can have a dimension **614A.sub.t** (e.g., thickness) along its Z-axis ranging from about 15 mm to about 25 mm (e.g., about 15 mm, about 20 mm, or about 25 mm). In some embodiments, base plate **614B** can have a dimension **614B.sub.w** (e.g., width) along an X-axis ranging from about 10 mm to about 50 mm (e.g., about 10 mm, about 20 mm, about 30 mm, or about 40 mm). The X-axes and Y-axes of

FIG. 6 can be along the diameters of cover plate **614A** and base plate **614B**. In some embodiments, base plate **614B** can have a dimension **614B.sub.t** (e.g., thickness) along a Z-axis ranging from about 1 mm to about 5 mm (e.g., about 1 mm, about 2 mm, about 3 mm, or about 4 mm).

(52) Cover plate **614A** is further described with FIGS. 7A-7B, which shows a back side view of cover plate **614A** and a cross-sectional view of cover plate **614A** along line A-A of FIG. 7A. Elements in FIGS. 7A-7B with the same annotations as elements in FIG. 6 are described above. Different regions of cover plate **614A** have been provided with different shading for illustration purposes. For example, the dotted regions represent recessed rim portions **672A-672B** and dark grey regions adjacent to dotted regions represent protruding rim portions **674A-674B** discussed above with reference to FIG. 6. Rim portions **672A** and **674A** can have arc shapes along outer circumference **614.sub.OC** of cover plate **614** and rim portions **674A** and **674B** can have arc shapes along inner circumference **614.sub.IC** of cover plate **614**.

(53) In some embodiments, an angle A ranging from about 20 degrees to about 30 degrees (e.g., about 20 degrees, about 24 degrees, or about 28 degrees) can be subtended by each of arc-shaped protruding rim portions **674A-674B** at center C of outer and inner circumferences **614.sub.OC** and **614.sub.IC**. In some embodiments, gas inlet port **662** can have a circular shape and a diameter line **662d** of circular-shaped gas inlet port **662** can form an angle B ranging from about 20 degrees to about 40 degrees (e.g., about 20 degrees, about 30 degrees, or about 40 degrees) with a diameter line of cover plate **614** along a Y-axis.

(54) FIG. 8 shows a back side view of a base plate **814B** of a gas distribution element with gas distribution holes **864**, according to some embodiments. The above discussion of base plate **614B** and gas distribution holes **664** applies to base plate **814B** and gas distribution holes **864**, respectively, unless mentioned otherwise. Base plate **814B** can have different regions with different sized gas distribution holes **864**. In some embodiments, base plate **814B** can have four regions with four sets **880A-880D** of gas distribution holes **864**. In some embodiments, each of four sets **880A-880D** can have gas distribution holes **864** with dimensions different from the other sets. In some embodiments, gas distribution holes **864** can each have dimensions substantially equal to or different from each other in the same set of set **880A**, **880B**, **880C**, and/or **880D**. In some embodiments, gas distribution holes **864** of set **880A** can have the smallest dimensions compared to sets **880B-880D** and gas distribution holes **864** of set **880C** can have the largest dimensions compared to sets **880A-880B** and **880D**.

(55) In some embodiments, gas distribution holes **864** of set **880A** can each have a diameter ranging from about 1 mm to about 5 mm (e.g., about 1 mm, about 2 mm, about 3 mm, or about 4 mm). In some embodiments, gas distribution holes **864** of set **880C** can each have a diameter ranging from about 8 mm to about 15 mm (e.g., about 8 mm, about 10 mm, about 12 mm, or about 15 mm). In some embodiments, gas distribution holes **864** of sets **880B** and **880D** can each have a diameter ranging from about 4 mm to about 8 mm (e.g., about 4 mm, about 6 mm, or about 8 mm). In some embodiments, gas distribution holes **864** of sets **880B** and **880D** can have dimensions equal to or different from each other. In some embodiments, the number of gas distribution holes **864** in each of sets **880A-880D** can be equal to or different from each other. In some embodiments, base plate **814B** can have a horizontal dimension **814B.sub.w** (e.g., width) along an X-axis ranging from about 30 mm to about 50 mm (e.g., about 30 mm, about 35 mm, about 40 mm, or about 45 mm).

(56) The arrangements and dimensions of gas distribution holes **864** in base plate **814B** can be configured to control the volume of thermally conditioned gas discharged from different regions of a gas distribution element (e.g., gas distribution elements **114A-114B**). The volume discharged from the different regions can be varied to compensate for variations in temperature of the thermally conditioned gas within the gas distribution element and consequently, provide a substantially uniform amount of thermal energy (hot or cold) from the discharged volume across a dielectric window (e.g., dielectric window **110**). The temperature of thermally conditioned gas in

regions of the gas distribution element near a gas inlet port (e.g., gas inlet port **662**) can be higher (e.g., heated gas) or lower (e.g., cooling gas) than that in regions of the gas distribution element farther away from the gas inlet port. This temperature variation can be due to heat loss or heat gain by the thermally conditioned gas when circulating within the gas distribution element.

(57) To compensate for the temperature variations and provide the substantially uniform amount of thermal energy across the dielectric window, the discharged volume of the thermally conditioned gas can be increased as the distance of the discharging region from the gas inlet port increases. The discharging region can be a region of the gas distribution element from where the thermally conditioned gas is discharged. The temperature variations can be compensated when the gas distribution element can have gas distribution holes **864** with the smallest dimensions (e.g., set **880A**) positioned under or closer to the gas inlet port compared to larger gas distribution holes **864** (e.g., sets **880B-880D**). In this position of base plate **814B** with respect to the gas inlet port, gas distribution holes **864** of sets **880B** and **880D** with larger dimensions can be farther away from the gas inlet port and gas distribution holes **864** of set **880C** with the largest dimensions can be farthest away from the gas inlet port. As a result, the smallest volume of the thermally conditioned gas can be discharged from gas distribution element regions near the gas inlet port, larger volumes can be discharged from regions farther away from the gas inlet port, and the largest volume can be discharged from regions farthest away from the gas inlet port. The variation in the dimensions of gas distribution holes **864** can depend on the variation of the thermally conditioned gas within the gas distribution element.

(58) FIG. **9** shows a back side view of a base plate **914B** of a gas distribution element with gas distribution holes **964**, according to some embodiments. The above discussion of base plate **614B** and gas distribution holes **664** applies to base plate **914B** and gas distribution holes **964**, respectively, unless mentioned otherwise. Similar to base plate **814B**, base plate **914B** can have different regions with different sized gas distribution holes **964**, but base plate **914B** can have a horizontal dimension **914B.sub.w** (e.g., width) along an X-axis smaller than horizontal dimension **814B.sub.w** (e.g., width) of base plate **814**. In addition, compared to base plate **814B**, base plate **914B** can have a different arrangement of gas distribution holes **964**.

(59) In some embodiments, base plate **914B** can have four regions with four sets **980A-980D** of gas distribution holes **964**. In some embodiments, each of four sets **980A-980D** can have gas distribution holes **964** with dimensions different from the other sets. In some embodiments, gas distribution holes **964** can each have dimensions substantially equal to or different from each other in the same set of set **980A**, **980B**, **980C**, and/or **980D**. In some embodiments, gas distribution holes **964** of set **980A** can have the smallest dimensions compared to sets **980B-980D** and gas distribution holes **964** of set **980C** can have the largest dimensions compared to sets **980A-980B** and **980D**.

(60) In some embodiments, gas distribution holes **964** of set **980A** can each have a diameter ranging from about 1 mm to about 5 mm (e.g., about 1 mm, about 2 mm, about 3 mm, or about 4 mm). In some embodiments, gas distribution holes **964** of set **980C** can each have a diameter ranging from about 8 mm to about 15 mm (e.g., about 8 mm, about 10 mm, about 12 mm, or about 15 mm). In some embodiments, gas distribution holes **964** of sets **980B** and **980D** can each have a diameter ranging from about 4 mm to about 8 mm (e.g., about 4 mm, about 6 mm, or about 8 mm). In some embodiments, gas distribution holes **964** of sets **980B** and **980D** can have dimensions equal to or different from each other. In some embodiments, the number of gas distribution holes **964** in each of sets **980A-980D** can be equal to or different from each other. In some embodiments, base plate **914B** can have horizontal dimension **914B.sub.w** (e.g., width) along an X-axis ranging from about 10 mm to about 30 mm (e.g., about 10 mm, about 15 mm, about 20 mm, or about 25 mm).

(61) Similar to base plate **814B**, the arrangements and dimensions of gas distribution holes **964** can be configured to control the volume of thermally conditioned gas discharged from different regions

of a gas distribution element (e.g., gas distribution elements **114A-114B**) to compensate for temperature variations in the thermally conditioned gas within the gas distribution element. The temperature variations can be compensated when the gas distribution element can have gas distribution holes **964** with the smallest dimensions (e.g., set **980A**) positioned under or closer to the gas inlet port compared to larger gas distribution holes **964** (e.g., sets **980B-980D**). In this position of base plate **914B** with respect to the gas inlet port, gas distribution holes **964** of sets **980B** and **980D** with larger dimensions can be farther away from the gas inlet port and gas distribution holes **964** of set **980C** with the largest dimensions can be farthest away from the gas inlet port. The variation in the dimensions of gas distribution holes **964** can depend on the variation of the thermally conditioned gas within the gas distribution element.

(62) Referring to FIGS. **8-9**, in some embodiments, base plate **814B** can be a base plate for inner gas distribution element **114A** and base plate **914B** can be a base plate for outer gas distribution element **114B** in plasma generation systems **104** and **404** of FIGS. **1** and **4**, respectively.

(63) FIG. **10** is a flow diagram of an example method **1000** for regulating and maintaining a temperature across a dielectric window of a temperature-controlled plasma generation system, according to some embodiments. This disclosure is not limited to this operational description. Rather, other operations are within the spirit and scope of the present disclosure. It is to be appreciated that additional operations may be performed. Moreover, not all operations may be needed to perform the disclosure provided herein. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **10**. In some implementations, one or more other operations may be performed in addition to or in place of the presently described operations. For illustrative purposes, method **1000** is described with reference to the embodiments of FIGS. **1**, **3A-3B**, **4**, **5A-5B**, **6**, **7A-7B**, and **8-9**. However, method **1000** is not limited to these embodiments.

(64) In operation **1005** of FIG. **10**, a plasma is generated for wafer plasma processing using a temperature-controlled plasma generation system. For example, as shown and discussed with reference to FIG. **1**, temperature-controlled plasma generation system **104** can be used to generate plasma within plasma processing chamber **102** for various plasma processes, such as etching or deposition on wafer **107**.

(65) In operation **1010** of FIG. **10**, a gas received by the temperature-controlled plasma generation system is thermally conditioned. For example, as shown and discussed with reference to FIG. **4**, thermal conditioning a gas (e.g., air or inert gas) received by temperature-controlled plasma generation system **404** can include receiving the gas by gas tube **550B** of gas conditioning system **116B**, directing the gas through gas tube **550A** using gas circulation system **554**, and heating or cooling the gas using thermal conditioning system **552**. In some embodiments, heating or cooling can include flowing the gas through a heating coil or a fan within gas tube **550A**, respectively. The gas can be thermally conditioned to a temperature ranging from about 70° C. to about 120° C. (e.g., about 70° C., about 80° C., about 90° C., about 100° C., about 110° C., or about 120° C.).

(66) In operation **1015** of FIG. **10**, the thermally conditioned gas is discharged on a dielectric window. For example, as shown and discussed with reference to FIG. **4**, discharging the thermally conditioned gas on dielectric window **110** can include directing the thermally conditioned gas to gas distribution element **114B** through gas inlet pipe **115B** using gas circulation system **554**. The discharging can further include discharging the thermally conditioned gas through gas distribution holes (e.g., holes **664** of FIG. **6**, holes **864** of FIG. **8**, or holes **964** of FIG. **9**) onto dielectric window **110** while circulating the thermally conditioned gas within gas distribution element **114B**. In some embodiments, gas circulation system **554** can include a blower for directing and circulating the thermally conditioned gas.

(67) In operation **1020** of FIG. **10**, the state of plasma generation is determined. For example, as shown and discussed with reference to FIG. **1**, based on the plasma generating gas supplied into plasma processing chamber **102** through gas injector **120**, control system **108** can output a control

signal (not shown) indicating whether plasma generation in plasma processing chamber **102** is terminated. If the output control signal indicates the termination of plasma generation, then method **1000** can proceed to operation **1025**, otherwise method **1000** can proceed to operation **1030**.

(68) In operation **1025** of FIG. **10**, the plasma processed wafer is removed from the plasma processing chamber in response to the plasma generation being terminated. For example, as shown and discussed with reference to FIG. **1**, wafer **107** can be removed from processing chamber **102** in response to the control signal from control system **108** indicating the termination of plasma generation in plasma processing chamber **102**.

(69) In operation **1030** of FIG. **10**, the temperature across the dielectric window is determined. For example, as shown and discussed with reference to FIG. **4**, determining the temperature across the dielectric window can include activating sensors **118A-118B** and/or **119** based on control signals (not shown) from control system **108**, measuring the temperatures using sensors **118A-118B** and/or **119**, and receiving the measured temperatures from sensors **118A-118B** and/or **119** by control system **108**.

(70) In operation **1035** of FIG. **10**, a difference between the determined temperatures and a desired temperature across the dielectric window is determined and the difference is compared to a baseline level. For example, as shown and discussed with reference to FIG. **1**, a processor (not shown) of control system **108** can compare a desired temperature set in the processor with the temperatures across dielectric window **110** received from sensors **118A-118B** and/or **119** and output a difference. The processor can compare the difference to a baseline level and output a signal indicating whether the difference is greater than, equal to, or less than the baseline level. The term “baseline level,” as used herein, refers to a temperature variation level that has been deemed to have minimal impact on wafer plasma processing or have any appreciable impact on wafer yield loss. In some embodiments, the baseline level can be one or more stored values in a database, a server, or on a local storage medium in processor **640c**.

(71) If the processor outputs a signal indicating the difference is greater than the baseline level, then method **1000** can proceed to operation **1040** followed by operation **1045**. In some embodiments, operations **1040** and **1045** can occur simultaneously if the processor outputs a signal indicating the difference is greater than the baseline level. Otherwise, if the processor outputs a signal indicating the difference is equal or below the baseline level, then method **1000** can proceed to operation **1045**.

(72) In operation **1040** of FIG. **10**, a parameter of a thermal conditioning system is adjusted in response to the difference between the determined temperatures and a desired temperature across the dielectric window being greater than the baseline level. For example, as shown and discussed with reference to FIGS. **1**, **4**, and **5A-5B**, control system **108** can dynamically adjust parameters of thermal conditioning system **552** to adjust the temperature of the thermally conditioned gas supplied to gas distribution elements **114A-114B**. The dynamic adjustment can be based on the determined temperatures across the dielectric window.

(73) In operation **1045** of FIG. **10**, a portion of the discharged thermally conditioned gas is extracted. For example, as shown and discussed with reference to FIG. **4**, extracting a portion of the discharged thermally conditioned gas can include drawing all or portions of the discharged thermally conditioned gas within chamber **442** through gas outlet port **444B** using gas circulation system **554**. In some embodiments, gas circulation system **554** can include a suction fan configured to draw the all or portions of the discharged thermally conditioned gas. After extraction, method **1000** can proceed to operation **1010** to thermally recondition and redistribute the extraction portion on dielectric window **110** through gas distribution elements **114A-114B**.

(74) The present disclosure provides example temperature-controlled plasma generation systems (e.g., plasma generation systems **104**, **304**, or **404**) in plasma processing systems (e.g., plasma processing system **100**) to prevent and/or mitigate the temperature dependent process drifts during plasma processing of a series of wafers to achieve plasma process repeatability and wafer-to-wafer

reproducibility. In some embodiments, a temperature-controlled plasma generation system can include a dielectric window (e.g., dielectric window **110**) and one or more gas distribution elements (e.g., gas distribution elements **114A**, **114B**, **314A**, or **314B**) positioned on the dielectric window. The one or more gas distribution elements can be configured to distribute thermally conditioned gas on the dielectric window and regulate the temperature across the dielectric window to be substantially uniform during wafer plasma processing.

(75) In some embodiments, the one or more gas distribution elements can have ring-shaped tubular structures and a plurality of gas distribution holes (e.g., gas distribution holes **664**, **864**, or **964**) configured to discharge the thermally conditioned gas onto a top surface of the dielectric window. In some embodiments, different regions of the one or more gas distribution elements can have different sets (e.g., sets **880A-880D** or **980A-980D**) of gas distribution holes. The dimension and density of each set of gas distribution holes can be different from each other. The sets of gas distribution holes can be strategically designed and arranged to control the thermally conditioned gas amount discharged across the dielectric window to provide substantially uniform amount of thermal energy (hot or cold) across the dielectric window and regulate the dielectric window temperature to be substantially uniform during wafer plasma processing.

(76) In some embodiments, a plasma generation system includes a dielectric window, an inductive coil disposed on the dielectric window, a gas distribution element disposed on the dielectric window, and a gas conditioning system coupled to the gas distribution element. The gas distribution element is configured to discharge a thermally conditioned gas on the dielectric window and regulate a temperature across the dielectric window. The gas conditioning system is configured to supply the thermally conditioned gas to the gas distribution element.

(77) In some embodiments, a plasma generation system includes a dielectric window, a gas injector coupled to the dielectric window, first and second inductive coils, first and second gas distribution elements, and a gas conditioning system coupled to the first and second gas distribution elements. The first and second inductive coils are disposed radially around the gas injector and on the dielectric window. The first and second gas distribution elements are disposed radially around the first and second inductive coils and on the dielectric window. Each of the first and second gas distribution elements is configured to discharge a thermally conditioned gas on the dielectric window. The gas conditioning system is configured to supply the thermally conditioned gas to the first and second gas distribution elements.

(78) In some embodiments, a method for regulating a temperature across a dielectric window of a plasma generation system includes thermally conditioning a gas supplied to the plasma generation system and discharging the thermally conditioned gas on the dielectric window. The method further includes determining a difference between a temperature across the dielectric window and a set temperature value and adjusting a parameter of the plasma generation system in response to the difference being greater than a baseline level.

(79) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for regulating a temperature across a dielectric window of a plasma generation system, the method comprising: thermal conditioning, using a gas conditioner, a gas supplied to the plasma

generation system to form a thermally conditioned gas; discharging, using a gas distributor, the thermally conditioned gas on the dielectric window; determining, using a controller, a difference between a temperature across the dielectric window and a predetermined temperature value; and adjusting, using the controller, a parameter of the plasma generation system in response to the difference being greater than a baseline level.

2. The method of claim 1, wherein thermally conditioning the gas comprises: receiving the gas through a gas tube of the gas conditioner; directing the gas through the gas tube using a gas circulator; and heating or cooling the gas using a thermal conditioner.

3. The method of claim 1, wherein discharging the thermally conditioned gas comprises directing the thermally conditioned gas to a gas distributor using a gas circulator.

4. The method of claim 1, wherein discharging the thermally conditioned gas comprises discharging the thermally conditioned gas through gas distribution holes in the gas distributor.

5. The method of claim 1, wherein discharging the thermally conditioned gas comprises discharging the thermally conditioned gas through first and second sets of gas distribution holes in the gas distributor, and wherein a number of the gas distribution holes in the first set is greater than a number of the gas distribution holes in the second set.

6. The method of claim 5, wherein a diameter of the gas distribution holes in the first set is smaller than a diameter of the gas distribution holes in the second set.

7. The method of claim 1, wherein adjusting the parameter of the plasma generation system comprises dynamically adjusting a parameter of a heating coil or a cooling fan of a thermal conditioner.

8. The method of claim 1, wherein determining the difference comprises: activating, using the controller, sensors disposed on the dielectric window; and measuring the temperature across the dielectric window using the sensors.

9. The method of claim 1, further comprising extracting a portion of the thermally conditioned gas after discharging on the dielectric window in response to the difference being equal to or less than the baseline level.

10. The method of claim 1, further comprising: extracting, using a gas circulator, a portion of the thermally conditioned gas after discharging on the dielectric window; and thermally reconditioning, using the gas conditioner, the portion of the thermally conditioned gas.

11. A method, comprising: conditioning, using a gas conditioner, a gas supplied to a processing chamber to form a conditioned gas; discharging, using a gas distributor, the conditioned gas on a dielectric window of the processing chamber; determining, using a controller, a difference between a temperature across the dielectric window and a predetermined temperature value; extracting, using a gas circulator, a portion of the conditioned gas after discharging on the dielectric window in response to the difference being equal to or less than a baseline level; and reconditioning, using the gas conditioner, the portion of the conditioned gas.

12. The method of claim 11, wherein discharging the conditioned gas comprises discharging different volumes of the conditioned gas on different regions of the dielectric window using the gas distributor.

13. The method of claim 11, wherein discharging the conditioned gas comprises: discharging a first volume of the conditioned gas through a first set of openings in the gas distributor and on a first region of the dielectric window; and discharging a second volume of the conditioned gas through a second set of openings in the gas distributor and on a second region of the dielectric window.

14. The method of claim 13, wherein a diameter of the openings in the first set of openings is smaller than a diameter of the openings in the second set of openings.

15. The method of claim 11, further comprising adjusting, using the gas conditioner, a temperature of the conditioned gas prior to discharging the conditioned gas in response to the difference being greater than the baseline level.

16. The method of claim 11, wherein determining the difference comprises: activating, using the

controller, sensors disposed on the dielectric window; and measuring the temperature across the dielectric window using the sensors.

17. A method, comprising: supplying, using a gas conditioner, a thermally conditioned gas to a gas distributor; discharging, using the gas distributor, the thermally conditioned gas on a window of a processing chamber; activating, using a controller, sensors disposed on the window and under the gas distributor; measuring, using the sensors, a temperature across the window; and adjusting, using the gas conditioner, a temperature of the thermally conditioned gas prior to discharging the thermally conditioned gas on the window.

18. The method of claim 17, wherein discharging the thermally conditioned gas comprises discharging different volumes of the thermally conditioned gas on different regions of the window.

19. The method of claim 17, wherein discharging the thermally conditioned gas comprises: discharging a first volume of the thermally conditioned gas on a first region of the window; and discharging a second volume of the thermally conditioned gas on a second region of the window.

20. The method of claim 17, wherein supplying the thermally conditioned gas comprises: receiving a gas through a gas tube of the gas conditioner; and heating or cooling the gas using a thermal conditioner.
